

TEMPORAL HAND LABANOTATION PRIORS FOR ROBUST CONTROL OF FROZEN VIDEO DIFFUSION

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ABSTRACT

Hand-controlled video generation requires reliable motion signals, yet current controllers assume clean joint sequences and reproduce tracking artifacts. We introduce HL-Repair, a compact temporal model that restores hand controls before a frozen video generator. HL-Repair combines continuous bone-direction features with soft Hand Labanotation (HL) posteriors pooled over neighboring frames. This representation regularizes abrupt directional changes while retaining continuous output compatible with the original controller. We train HL-Repair on corrupted STB trajectories and evaluate it against fixed filters and learned pose refiners on held-out sequences. Across 1,000 Wan2.1-Fun-Control videos with matched prompts and noise, restoration raises clean-output SSIM from .713 to .884 and exceeds median filtering by .035. The HL branch also improves trajectory restoration over a continuous-only variant. The model therefore strengthens hand control without modifying or retraining the video generator.

Index Terms— hand motion, controllable video, Hand Labanotation, temporal restoration, robustness

1. INTRODUCTION

Fine hand articulation is central to interactive video generation, but it is also unusually fragile. Recent systems improve hand control through joint-level conditioning, explicit 3-D geometry, and occlusion-aware feature propagation [1, 2, 3]. Their control signals nevertheless originate from a tracker, while evaluation commonly assumes those signals are clean. In practice, occlusion may freeze a fingertip, displace a joint, or create a one-frame outlier. A powerful generator can render such errors convincingly rather than correct them, and retraining a billion-parameter model for every tracker or capture domain is impractical.

We therefore ask whether a compact, generator-agnostic module can repair corrupted hand trajectories before they reach a frozen controller. Ordinary low-pass filters suppress jitter, but they also blur genuine articulation and cannot distinguish a meaningful direction change from an implausible excursion. Hand Labanotation (HL) provides the missing

structure: its 26 spatial symbols organize hand directions into a human-interpretable vocabulary [4]. HL-Repair converts noisy directions into soft HL evidence, pools that evidence over time, and fuses it with continuous motion features. The repaired joints remain continuous and follow the unmodified Wan2.1-Fun-Control interface [5, 6].

Our contributions are threefold: (1) a compact temporal front end that repairs tracker-corrupted controls while leaving the video generator unchanged; (2) a soft, temporally pooled HL representation that contributes at inference, rather than serving only as auxiliary supervision; and (3) a paired evaluation spanning held-out 3-D motion and 1,000 videos generated by a frozen Wan controller under matched prompts and noise.

2. RELATED WORK

Hand-controlled video. Existing systems condition generation on joints, 3-D geometry, free-space gestures, or point clouds [1, 3, 2, 7]. Others modify diffusion sampling from coarse reference animations [8], while ControlNet and LoRA adapt the generator itself [9, 10]. These approaches improve how motion is consumed, but generally assume that the incoming control is trustworthy. We instead target the complementary upstream problem: restoring stale, mistracked, or outlying hand joints before an unchanged native controller.

Motion restoration and representation. SmoothNet learns a temporal refiner for noisy human poses [11], and MTVCraft learns motion tokens jointly with an animation system [12]. Our setting differs in both representation and deployment: HL supplies a fixed, interpretable directional vocabulary, and the repaired trajectory is evaluated through a frozen downstream generator. Continuous reconstruction preserves fine geometry, while soft HL posteriors provide semantic evidence at inference; only the auxiliary output classifier is removed after training.

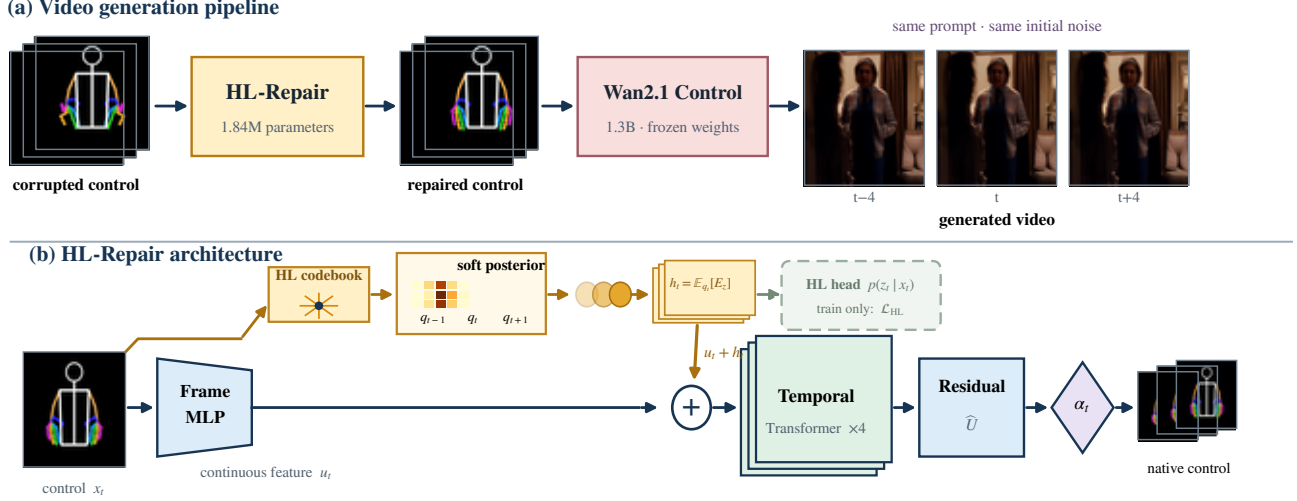


Fig. 1. HL-Repair changes only the tracker-side signal. Soft HL posteriors are pooled locally and injected into the temporal repairer; clean symbols also supervise training. A gate bypasses trustworthy clips, and restored joints use Wan’s native control interface.

3. METHOD

3.1. Canonical hand stream

Let $p_{t,k} \in \mathbb{R}^3$ be joint k at frame t , with wrist $k = 0$ and parent $\pi(k)$. Each bone direction is

$$u_{t,k} = p_{t,k} - p_{t,\pi(k)} / \|p_{t,k} - p_{t,\pi(k)}\|_2. \quad (1)$$

Directions are expressed in a right-handed palm frame: its distal axis points from wrist to middle MCP, its normal is the cross product of the index-to-little MCP axis with the distal axis, and the third axis completes the frame. This removes global translation, scale, and proper rotation. A subject’s calibrated bone lengths and global carrier are retained as side information, so restoration changes articulation only. We use the HL regional-vector tree rather than a conventional five-ray palm: wrist-to-middle MCP anchors the palm, with transverse middle-to-index, middle-to-ring, and ring-to-little links; phalangeal chains are unchanged [4].

We simulate three tracker failures. Isotropic direction jitter has per-clip standard deviation sampled uniformly from .03 to .10 before renormalization. Independently, 10% of bone observations are replaced by random unit directions, modeling directional outliers, and 8% are replaced by the preceding observation, modeling a stale tracker. The same realized corruption is supplied to every method. Training uses eight equally weighted states: clean input; isolated jitter, directional outliers, and stale observations; mild mixed corruption; two nominal mixed realizations; and severe mixed corruption. This teaches restoration without assuming that every incoming clip is equally damaged.

3.2. HL-supervised temporal repair

HL-Repair consumes a 17-frame tensor $\tilde{U} \in \mathbb{R}^{17 \times 20 \times 3}$. A linear layer maps each flattened frame to width 192; four pre-normalized Transformer encoder blocks with six heads model the sequence [13]. A residual head gives

$$\hat{u}_{t,k} = \text{norm}(\tilde{u}_{t,k} + g_\theta(\tilde{U})_{t,k}). \quad (2)$$

The training model contains 1.94M parameters; discarding its 100,360-parameter output classifier leaves 1.84M at deployment. A repairer forward pass for one 17-frame window takes 2.43 ms median on one CPU thread (300 timed runs after 30 warm-ups). Zero-initialized residual weights make the network implement the identity mapping at initialization.

For HL supervision, define the reproducible 26-center set

$$C = \{a / \|a\|_2 : a \in \{-1, 0, 1\}^3 \setminus \{0\}\}, \quad (3)$$

$$q(u) = \arg \max_{c \in C} c^\top u. \quad (4)$$

We instantiate HL’s 26-part cuboid construction as rays from the center of an equal-axis cuboid to its six face centers, 12 edge midpoints, and eight vertices [4]. Rather than feed brittle hard symbols, we form $r_{t,k} = \text{softmax}(C\hat{u}_{t,k} / .24)$, average r over three adjacent frames, embed its expectation, and add the resulting 20-symbol feature to the continuous frame embedding. Thus HL participates in inference while the output remains continuous. A second head emits logits z for clean $q(u_{t,k})$. For $s = \text{softmax}(z)$ and target center c_y , we define $\mathcal{L}_{HL} = \text{CE}(z, y) + \sum_j s_j (1 - c_y^\top c_j)$; the second term penalizes predictions that are angularly distant from the target symbol.

Training minimizes

$$\mathcal{L} = \mathbb{E}[1 - \hat{u}^T u] + .2 \text{SmoothL1}(\Delta \hat{U}, \Delta U) + .003 \mathcal{L}_{HL}(h_\theta, q(U)). \quad (5)$$

The continuous-only ablation removes the HL posterior branch and loss. To separate semantics from local context or capacity, a 1.94M-parameter three-frame continuous-context baseline projects concatenated previous, current, and next directions; it is slightly larger than HL-Repair.

Repair should not alter a trustworthy stream. Without requiring tracker-specific confidence, we compute each clip’s mean angular deviation from its centered median-9 trajectory and apply the network only above 3.75° . We select this threshold on B6 to balance restoration accuracy and clean-input preservation, and keep it fixed for B1.

4. EXPERIMENTS

4.1. Protocol

Real motion and split. We use official STB 3-D annotations [14] and exclude the synchronized second camera. Hyperparameters are selected on B6; final models are then trained on the ten Counting and Random sequences from B2–B6 (15,000 frames) and evaluated once on the disjoint B1 sequences (3,000 frames). Training uses 3,710 stride-four windows and the eight-state curriculum (29,680 samples), while testing uses 176 non-overlapping windows under a fixed unseen corruption. The selected configuration uses an HL-loss weight of .003, three-frame posterior pooling, and temperature .24. We train for 40 epochs with AdamW at 3×10^{-4} on an Ascend 910B2C NPU; five independent seeds evaluate the HL representation.

Baselines and metrics. We compare the corrupted stream, causal EMA, causal and centered median filters, Savitzky–Golay filtering, SmoothNet [11], and the identical Transformer without $\mathcal{L}_{HL}$. SmoothNet is retrained on our split with 17-frame inputs; we report its original-width and capacity-matched 1.86M variants. Filter hyperparameters are selected on B6 and fixed for B1. We measure geodesic direction error, exact HL-symbol accuracy, and acceleration error $\|\Delta^2 \hat{U} - \Delta^2 U\|_2$. Paired 95% intervals resample the 176 test windows 10,000 times.

Frozen Wan. Fifty evenly spaced held-out STB clips are decoded with calibrated lengths and placed on a common upper-body carrier. We compare clean, corrupted, median-9, continuous-only, and HL-repaired controls. Four diffusion seeds produce 1,000 text-conditioned, 17-frame, 256^2 videos with Wan2.1-Fun-V1.1-1.3B-Control; prompts, model settings, and initial noise are matched within each comparison. DWPose [15] measures hand-keypoint accuracy, while frame-wise RGB SSIM [16] measures similarity to the same-seed clean-control video. HL-Repair versus median-9 is the primary comparison. Because detector coverage varies across

Table 1. Held-out STB restoration: angular error (degrees), exact HL accuracy (%), and direction-acceleration error.

Method	Angle ↓	Acc. (%) ↑	Accel. ↓
Corrupted	13.24	82.79	.600
Causal EMA	11.37	78.52	.272
Causal median-3	8.01	86.98	.216
Centered median-9	4.26	91.26	.105
Savitzky–Golay	6.86	85.61	.0806
SmoothNet (1.86M)	3.61	92.40	.0847
Continuous-only	2.99	92.45	.0840
HL-Repair	2.92	92.74	.0826

frames, we additionally report padded hand-region SSIM and treat DWPose scores as diagnostic. Statistics are computed over trajectories after averaging their four seeds; paired significance uses a Monte Carlo sign-flip test.

4.2. Results

Table 1 shows that learned refiners outperform fixed filters in angular error, with HL-Repair attaining the lowest error and highest symbol accuracy. It reduces mean error from 13.24° to 2.92° , a 31.5% gain over median-9; joint MPJPE likewise falls from 15.18 to 2.10 mm (median: 3.53 mm). Against capacity-matched 1.86M-parameter SmoothNet, HL-Repair reduces angle error by $.70^\circ$ and MPJPE by $.72$ mm [.63, .81]. This advantage holds for all five matched training seeds, averaging $.700^\circ$ [.681, .718].

Compared with the continuous-only model, HL improves mean angle by $.071^\circ$ [.041, .102] and transition F1 by 4.36 points [2.41, 6.22]. Across five trainings, mean angle improves by $.068^\circ$ (seed-level 95% CI [.022, .114]), P95 by $.241^\circ$ [.023, .460], and symbol accuracy by .307 points [.146, .467]. Soft posteriors are important: hard symbols yield 3.418° on B6, while the continuous local-context control obtains 3.207° , compared with 3.151° for pooled HL. Across corruption draws, the angular gain is $.075 \pm .011^\circ$, 5/5, and increases under severe corruption.

The learned repairer is paired with a conservative gate. It activates on every nominally corrupted B1 window but on only 0.6% of clean windows, reducing clean-input error from 1.82° under unconditional repair to $.015^\circ$.

Table 2 shows that trajectory repair remains effective after the frozen generator. HL-Repair lowers rendered-joint error from 5.42 to .65 pixels (median-9: 1.17) and raises same-seed clean-output SSIM from .713 to .884. Relative to median-9, the SSIM gain is .035 [.020, .049] with 41/50 wins ($p < 10^{-6}$), while LPIPS changes by $-.031$ [-.043, -.019].

HL and the continuous-only model produce comparable videos ($\Delta\text{SSIM} = .000$ [-.006, .006]; $\Delta\text{LPIPS} = -.0004$ [-.0060, .0052]), indicating that the downstream gain comes primarily from temporal restoration. Qualitatively, restoration recovers the clean-control composition despite matched-noise

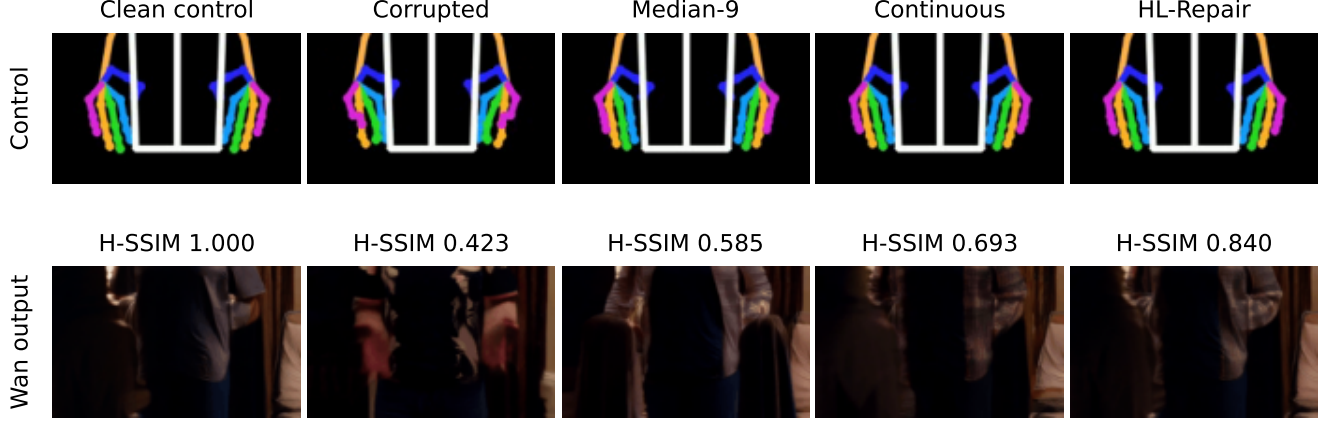


Fig. 2. Qualitative comparison under matched prompt and noise. HL-Repair restores the hand configuration and better preserves the clean-control appearance.

Table 2. Frozen-Wan paired evaluation. Ctrl denotes rendered-joint error in pixels; image metrics use the same-seed clean output. H-SSIM and LPIPS are supplementary diagnostics.

Control	Ctrl ↓	SSIM ↑	H-SSIM ↑	LPIPS ↓
Clean oracle	0.00	1.000	1.000	.000
Corrupted	5.42	.713	.600	.221
Centered median-9	1.17	.850	.776	.115
Continuous-only	.67	.884	.832	.084
HL-Repair	.65	.884	.831	.084

changes in identity and clothing (Fig. 2).

Robustness. For B1Counting, the gain over median-9 is .032; for B1Random, it is .037, and deleting any one trajectory leaves a positive interval. A post-hoc check on two additional prompts gives HL-minus-median SSIM of .053 [−.010, .115] for chef and .012 [−.025, .050] for office; both intervals cross zero.

5. LIMITATIONS AND CONCLUSION

HL-Repair uses bidirectional 17-frame context and is therefore not streaming. The current evaluation assumes a reliable palm carrier and calibrated bone lengths; STB contains one subject and no object contact, and the downstream study uses one 1.3B backbone. Future work should evaluate streaming repair on multi-subject interaction data and larger video generators. Although HL contributes less than temporal restoration itself, its gain is consistent across training seeds and increases under severe corruption. HL-Repair thus addresses a practical weakness of hand-controlled generation without generator fine-tuning.

6. COMPLIANCE WITH ETHICAL STANDARDS

This computational study uses only publicly released STB 3-D annotations and machine-generated videos. It recruited no participants, collected no new human-subject data, and accessed no identifiable source imagery. OpenAI Codex (GPT-5) assisted literature synthesis, study design, writing, code, experiment orchestration, and reproducibility preparation; the authors remain responsible.

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